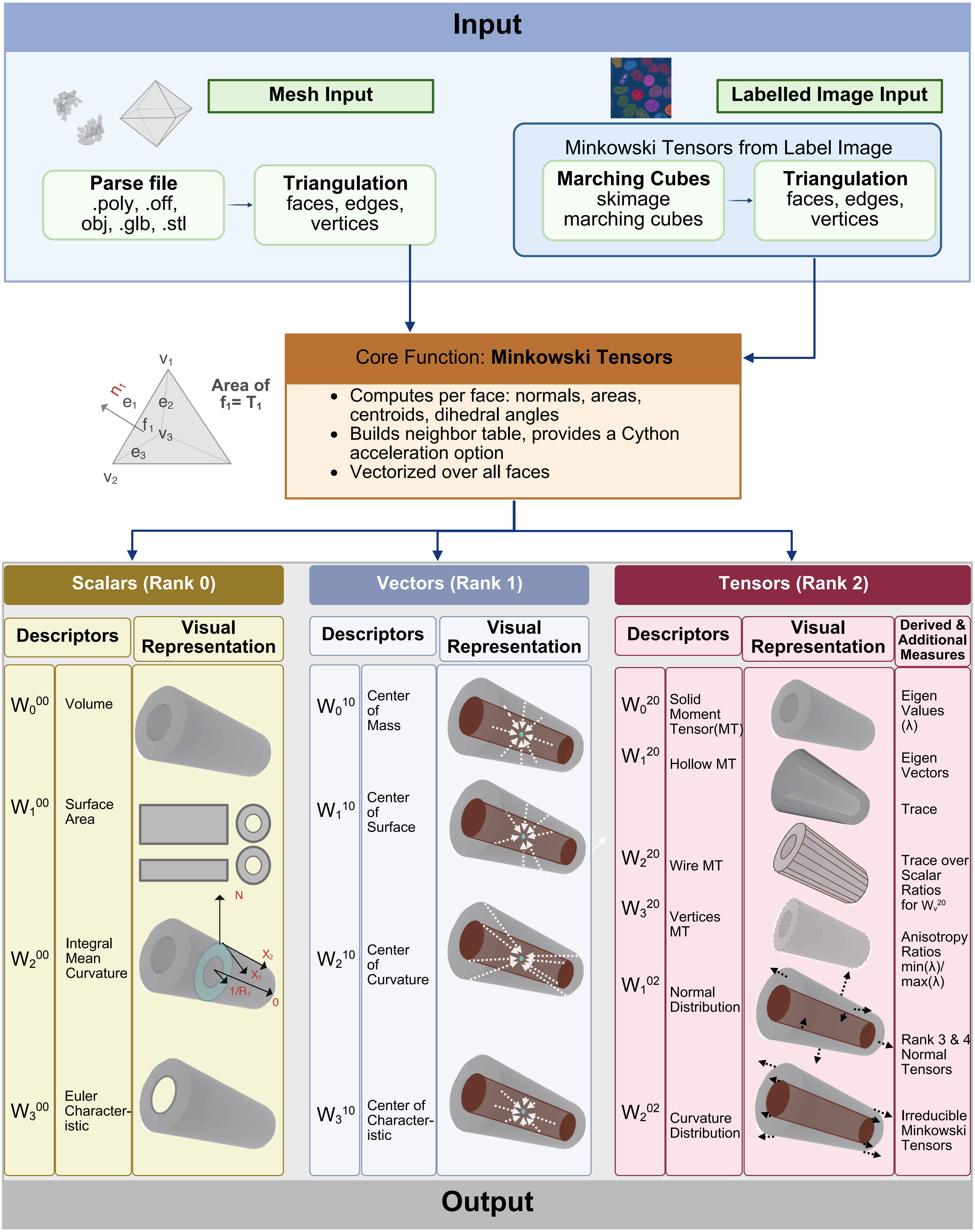

Input
Mesh Input
Labelled Image Input
Parse file
.poly, .off,
obj, .glb, .stl
Triangulation
faces, edges,
vertices
Minkowski Tensors from Label Image
Marching Cubes
skimage
marching cubes
Triangulation
faces, edges,
vertices
v_1
n_1
e_1
e_2
f_1
v_3
e_3
v_2
Area of $f_1 = T_1$
Core Function: **Minkowski Tensors**
• Computes per face: normals, areas, centroids, dihedral angles
• Builds neighbor table, provides a Cython acceleration option
• Vectorized over all faces
Scalars (Rank 0)
Vectors (Rank 1)
Tensors (Rank 2)
Descriptors
Visual Representation
Derived & Additional Measures
W_0^{00} Volume
W_1^{00} Surface Area
W_2^{00} Integral Mean Curvature
N
X_2
X_1
$1/R_1$
0
W_3^{00} Euler Character-istic
W_0^{10} Center of Mass
W_1^{10} Center of Surface
W_2^{10} Center of Curvature
W_3^{10} Center of Character-istic
W_0^{20} Solid Moment Tensor(MT)
W_1^{20} Hollow MT
W_2^{20} Wire MT
W_3^{20} Vertices MT
W_1^{02} Normal Distribution
W_2^{02} Curvature Distribution
Eigen Values (λ)
Eigen Vectors
Trace
Trace over Scalar Ratios for W_ν^{20}
Anisotropy Ratios min(λ)/max(λ)
Rank 3 & 4 Normal Tensors
Irreducible Minkowski Tensors
Output